

Project 2: Questions for Image Classification (CNN)

Answer these by implementing the CNN solution and, where asked, with short written answers in your notebook or report.

Section A: Data and Preprocessing

1. Q1. What is the shape of a single image in your dataset (height, width, channels)? How many training and test samples are there? How many classes?
2. Q2. How did you normalize pixel values? Why is normalization important for training a CNN?
3. Q3. Did you use a validation split? If yes, how did you create it (e.g., from training data)? What is the rationale for having a separate validation set?
4. Q4. Describe the data augmentation techniques you applied (e.g., rotation, horizontal flip, zoom). For each, state the parameter range (e.g., rotation range in degrees) and why it is suitable for your dataset.

Section B: Model Architecture and Training

5. Q5. Describe your CNN architecture layer by layer: number of conv layers, filter sizes, pooling, flatten, dense layers, and activation functions. How many trainable parameters does your model have?
6. Q6. Why do we use convolutional layers instead of only dense layers for image data? Explain in terms of spatial structure and parameter count.
7. Q7. Where did you apply dropout (after which layers) and what dropout rate did you use? How did dropout affect validation accuracy or overfitting compared to a model without dropout?
8. Q8. Did you use early stopping? If yes, what metric did you monitor and what patience value did you set? If not, how did you choose the number of epochs?

Section C: Evaluation and Interpretation

9. Q9. Report test set accuracy and display the confusion matrix. Which class(es) had the lowest accuracy? Suggest one possible reason (e.g., visual similarity between classes).
10. Q10. Plot (or describe) training vs validation loss and accuracy. Did the model overfit? How did data augmentation and dropout change the gap between training and validation performance?

Section D: Extension and Reflection

11. Q11. If you had more compute and time, name one architectural change you would try (e.g., more layers, batch normalization, different optimizer) and why.
12. Q12. In a real product (e.g., mobile app), what would you consider for deploying this CNN (e.g., model size, latency, quantization)?